

MediSync AI

Industry Standard HMS vs. MediSync AI Architectural Comparison & Phased Roadmap

Feature Coverage Analysis, Deployment Blueprint, and Enterprise Evolution Strategy

Document Version: 1.0.0	Target Architecture: Enterprise HMS
Scope: Full System Analysis & Roadmap	Classification: Executive Overview

1. Industry Standard HMS vs. MediSync AI Comparison

Module / Domain	Industry Standard HMS	MediSync AI Implementation	Coverage
Outpatient (OPD)	Walk-in intake, online booking, token queues, triage	Full booking desk, intake forms, priority token queue	100%
Inpatient (IPD)	Admissions, ward/bed management, discharge summary	Real-time Bed & Ward Occupancy dashboard across wards	90%
EHR Records	Diagnostic history, vitals, allergy logs, clinical notes	Patient EHR vault, Vitals Tracker (BP, HR, Temp, SpO2)	95%
e-Prescribing & Pharmacy	Digital RX, stock alerts, prescription dispensing queue	Digital RX, doctor digital signature, dispensing queue	90%
Revenue Cycle (RCM)	Itemized invoices, online payments, financial analytics	Itemized billing, online payments, Recharts analytics	90%
Telemedicine	HD video call rooms, live chat, remote biometrics	Virtual telehealth video calls, live chat, vitals sync	95%
Role Access (RBAC)	Multi-role permissions for Admin, Doctor, Receptionist	JWT auth for Admin, Doctor, Receptionist, Patient, Pharmacy	100%
AI Support	AI symptom triage, differential diagnosis hints	MediSync AI Assistant for symptom triage & summaries	95%
LIS / RIS Diagnostics	DICOM viewer, lab sample tracking, analyzer sync	Scanned attachments, diagnostic report image logs	60%
Infrastructure	PostgreSQL DB, Docker containers, REST API, HTTPS	PostgreSQL + SQLite fallback, Express REST API, Docker	95%

2. Phase-Wise Development & Deployment Roadmap

Phase	Milestone Title	Status	Key Output
Phase 1	Core HMS Setup & Documentation	Completed	Full web app suite, REST API, AI features, PDF/MD manuals

Phase 2	VPS Production Deployment	In Progress	PostgreSQL Docker deployment, SSL, environment secrets
Phase 3	LIS/RIS DICOM & Insurance	Planned	DICOM image viewer, FHIR standards, automated insurance claims
Phase 4	Mobile Apps & Health IoT	Planned	Native iOS/Android apps, wearable biometrics, predictive AI

Phase Details & Execution Deliverables:

- **Phase 1 (Completed):** Core OPD, IPD, EHR, Pharmacy, Telehealth, and AI clinical assistant workflows built and documented with 26 GUI screenshots.
- **Phase 2 (In Progress):** Provisioning VPS instance (Ubuntu 22.04), Docker Compose stack with PostgreSQL, Nginx reverse proxy, SSL, and automated DB backups.
- **Phase 3 (Near-Term):** FHIR/HL7 interoperability, DICOM viewer for X-Rays/MRI scans, electronic claims clearinghouse, and SMS/WhatsApp reminders.
- **Phase 4 (Long-Term):** iOS/Android native mobile apps, Bluetooth health wearable biometric sync, multi-branch hospital chain management, and ER surge prediction.